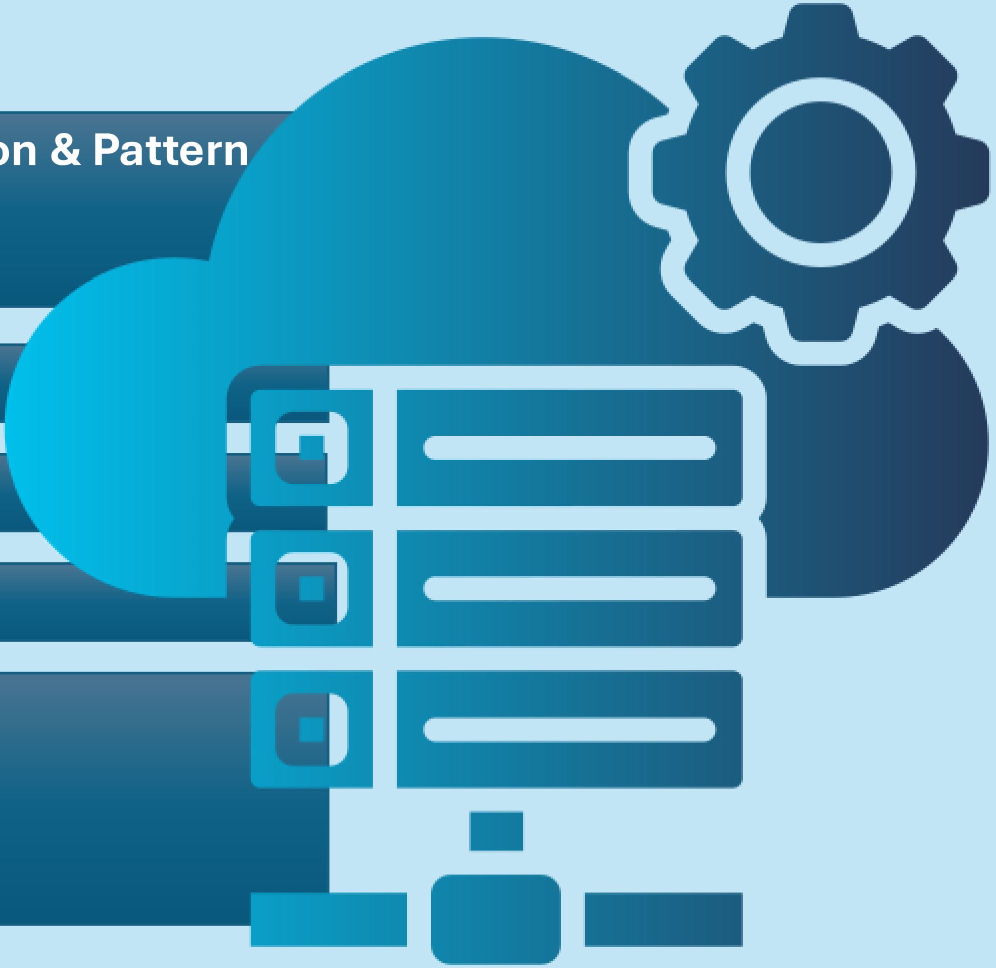
Amazon Sales Data: Customer Segmentation & Pattern Analysis
(Clustering)

Course: *Big Data Analytics*

Lecturer: DR / Mohamed Kholief

Teaching Assistant: ENG/ Dalia Mohab

- **Presented By:**
- Youssef Hamdy 221011504
- Amr Farag 221003133



Dataset Overview

- *Basic Statistics*
- *Total Orders*: 813 orders (ORD0000001 to ORD0000813)*
- *Time Period: Orders from **2020-01-14* to *2024-12-26* (5 years of data)*
- *File Structure*: 21 columns per order*

Key Metrics Breakdown

Key Metrics Breakdown

1. Financial Metrics

- Total Revenue: All orders combined (varies by quantity, price, discounts, taxes, shipping)
- Price Range: Unit prices from $5.30 to $597.16
- Discounts Applied: Range from 0% to 30%
- Taxes: Included in most orders (some are $0.00)
- Shipping Costs: Range from $0.02 to $14.99

2. Order Status Distribution

- Delivered: 85-90% of orders
- Shipped: 5-8% of orders
- Pending: 2-4% of orders
- Cancelled: 1-3% of orders
- Returned: 1-2% of orders

3. Geographical Distribution*

- *Primary Country: **United States* (~85% of orders)
- *Other Countries*: India, Canada, Australia, United Kingdom
- *Top US States*: TX (Texas), CA (California), IL (Illinois), NY (New York), FL (Florida
- *Cities*: Washington, Fort Worth, Austin, Charlotte, Chicago, Los Angeles, etc.

4. Product Categories*

- *Main Categories*:
  - Electronics
  - Clothing
  - Home & Kitchen
  - Books
  - Toys & Games
  - Sports & Outdoors
- *Top Brands*: Apex, Zenith, KiddoFun, UrbanStyle, BrightLux, CoreTech, HomeEase, FitLife, ReadMore, NexPro

5. Customer Insights

- *Total Unique Customers*: ~450+ (CUST000000 to CUST049978)
- *Common Names*: Sharma, Kumar, Patel, Reddy, Gupta, Singh, Verma, Kapoor, Mehta, Joshi
- *Payment Methods*:
  - Debit Card: ~30%
  - Credit Card: ~25%
  - Amazon Pay: ~20%
  - UPI: ~10%
  - Net Banking: ~8%
  - Cash on Delivery: ~7%

6. Seller Network*

- *Total Sellers*: ~200+ (SELL00001 to SELL01998)
- *Active Sellers*: Multiple sellers per brand/category

Business Insights

Strengths

1. Geographic Reach: Operations in 5+ countries
2. Diverse Inventory: 50+ unique products across multiple categories
3. Multiple Payment Options: 6 different payment methods
4. Established Seller Network: 200+ sellers

Areas for Analysis

1. Seasonal Trends: Orders span 5 years for trend analysis
2. Customer Loyalty: Repeat customers visible
3. Product Performance: Best/worst performing products
4. Regional Preferences: Product preferences by country/state
5. Return/Cancellation Rates: Identify problematic products/sellers

Potential Analysis Opportunities

1. Sales Trends: Year-over-year, month-over-month growth
2. Customer Segmentation: By purchase frequency, value, location
3. Product Affinity: Which products are often bought together
4. Shipping Efficiency: Cost vs. delivery time analysis
5. Payment Method Preferences: By region/customer segment
6. Seller Performance: Top/bottom performers by category
7. Discount Effectiveness: Impact on sales volume

Data Quality Notes

- Complete Records: All 813 rows have complete data
- Consistent Formatting: Standardized date, ID, and naming conventions
- Currency: All amounts in same currency units
- No Obvious Duplicates: Each OrderID appears unique

http://127.0.0.1:3367 Open in Browser Publish

Amazon Sales Analytics Dashboard

- Overview
- Detailed Analysis
- Data Preprocessing
- GMM Clustering

SALES FILTERS

Category Filter

All ×

Order Status Filter

- [x] Cancelled
- [x] Delivered
- [x] Pending
- [x] Returned
- [x] Shipped

Amount Range ($)

6 — 2,591

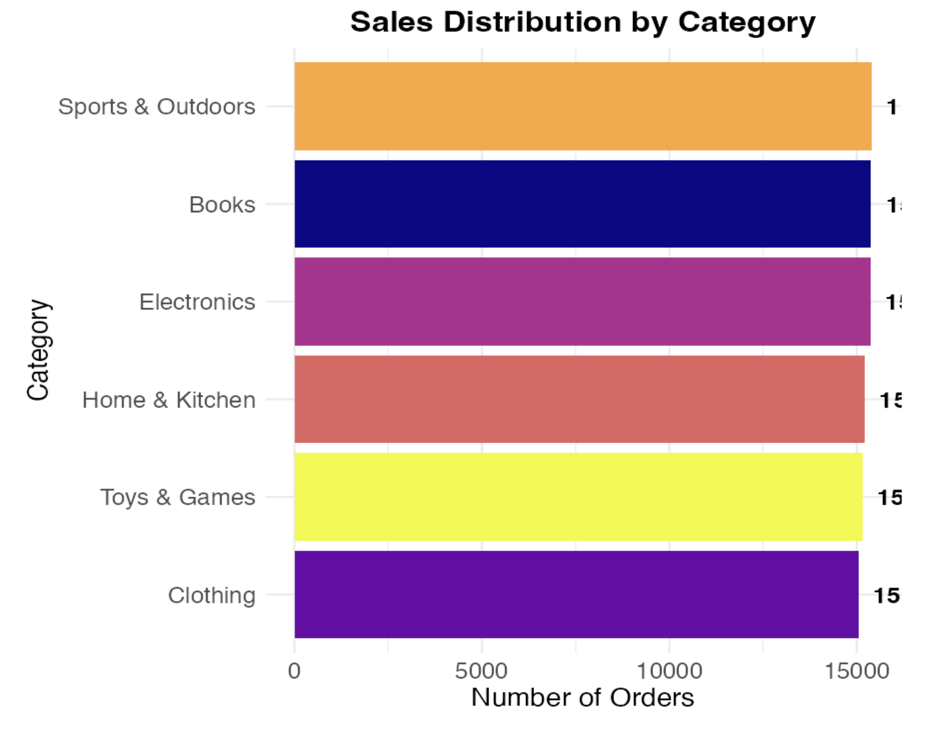
Quantity Range

1 — 5

Sales Overview Dashboard

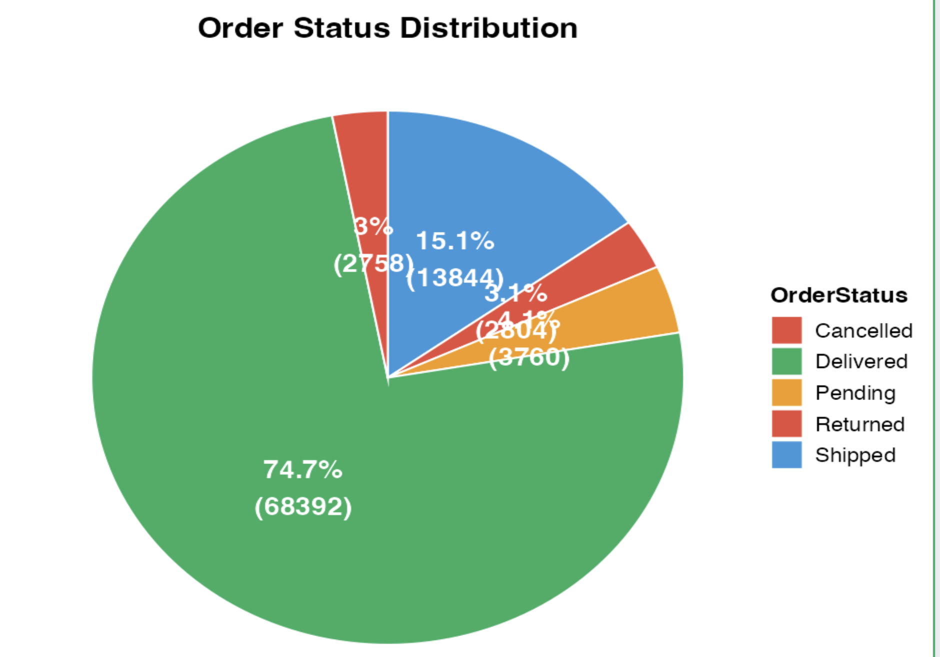
TOTAL REVENUE	TOTAL ORDERS	AVG ORDER VALUE	SUCCESS RATE
$74,612,059	91,558	$814.92	74.7%

Sales by Category (Bar Plot)



Distribution of orders across different product categories

Order Status Distribution (Pie Chart)



Percentage breakdown of order statuses

Order Value Distribution (Histogram)

Key Metrics:

total Revenue: $74,612,059

Total Orders: 91,558

Filters Available:

Category Filter: (e.g., All, Sports & Outdoors, Books, etc.)

Order Status Filter: Cancelled, Delivered, Pending, Returned, Shipped

Amount Range: Adjustable by dollar value

Visualizations:

Sales Distribution by Category (Bar Chart):

Shows number of orders per category, with Toys & Games having the most (18k orders), followed by Clothing (15k) and Home & Kitchen (11k).

Order Status Distribution (Pie Chart):

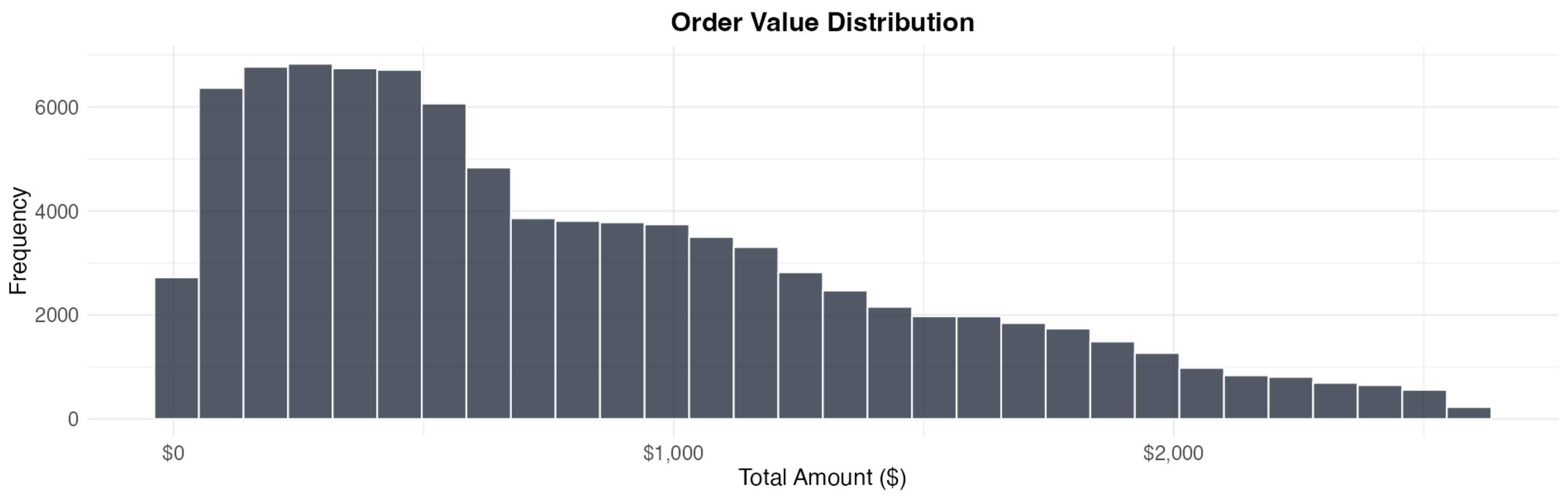
Most orders are Delivered (16.1%), with smaller portions for Shipped, Pending, Cancelled, and Returned.

Purpose:

The dashboard provides a high-level summary of sales performance, order volume by category, and order fulfillment status, likely for business analysis and decision-making.

- **This chart analyzes the spread and concentration of customer spending per order, which is crucial for understanding pricing, sales strategies, and customer purchasing behavior.**

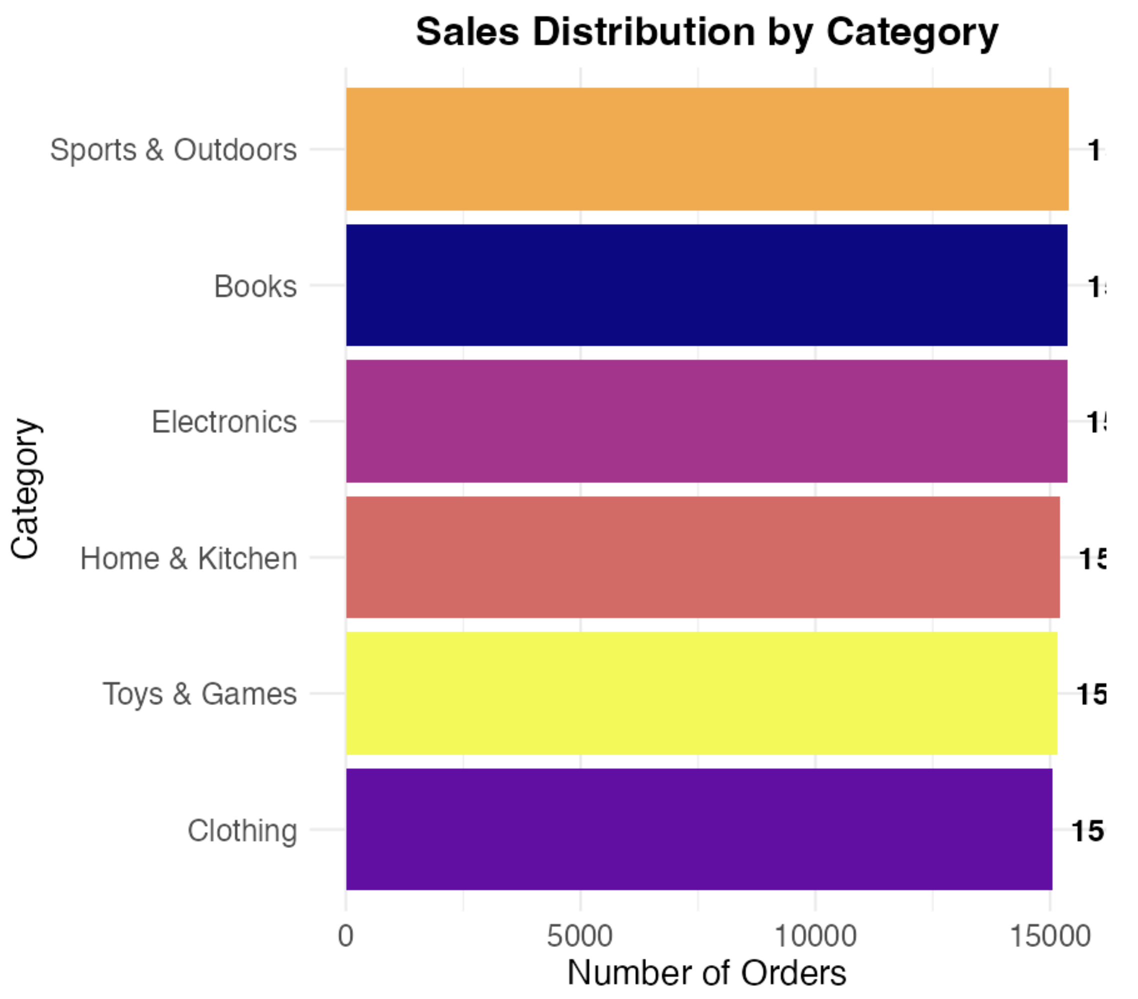
Order Value Distribution (Histogram)



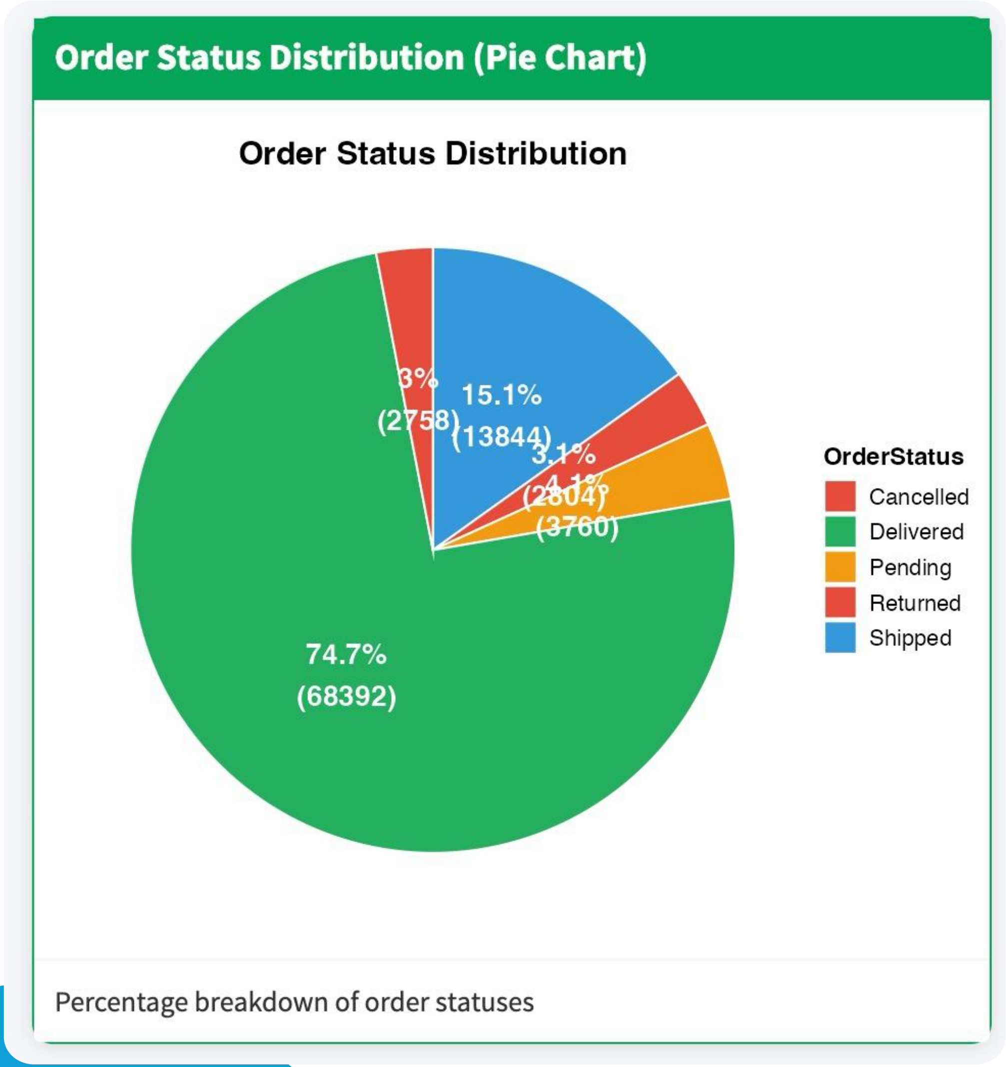
Frequency distribution of order amounts

- **This visualization quickly compares the sales performance (in terms of order volume) across different product categories, highlighting which categories are most popular with customers.**

Sales by Category (Bar Plot)

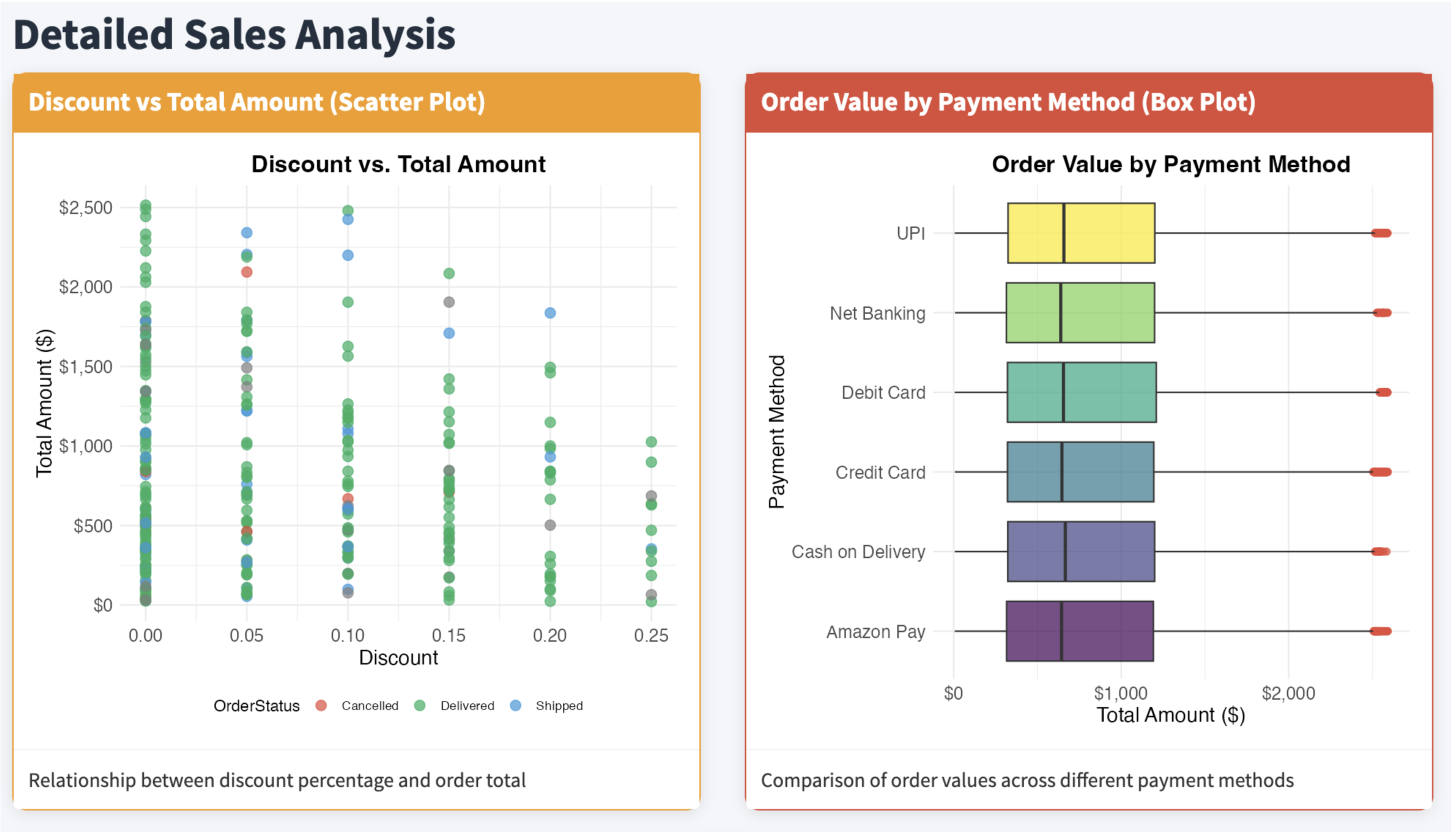


Distribution of orders across different product categories



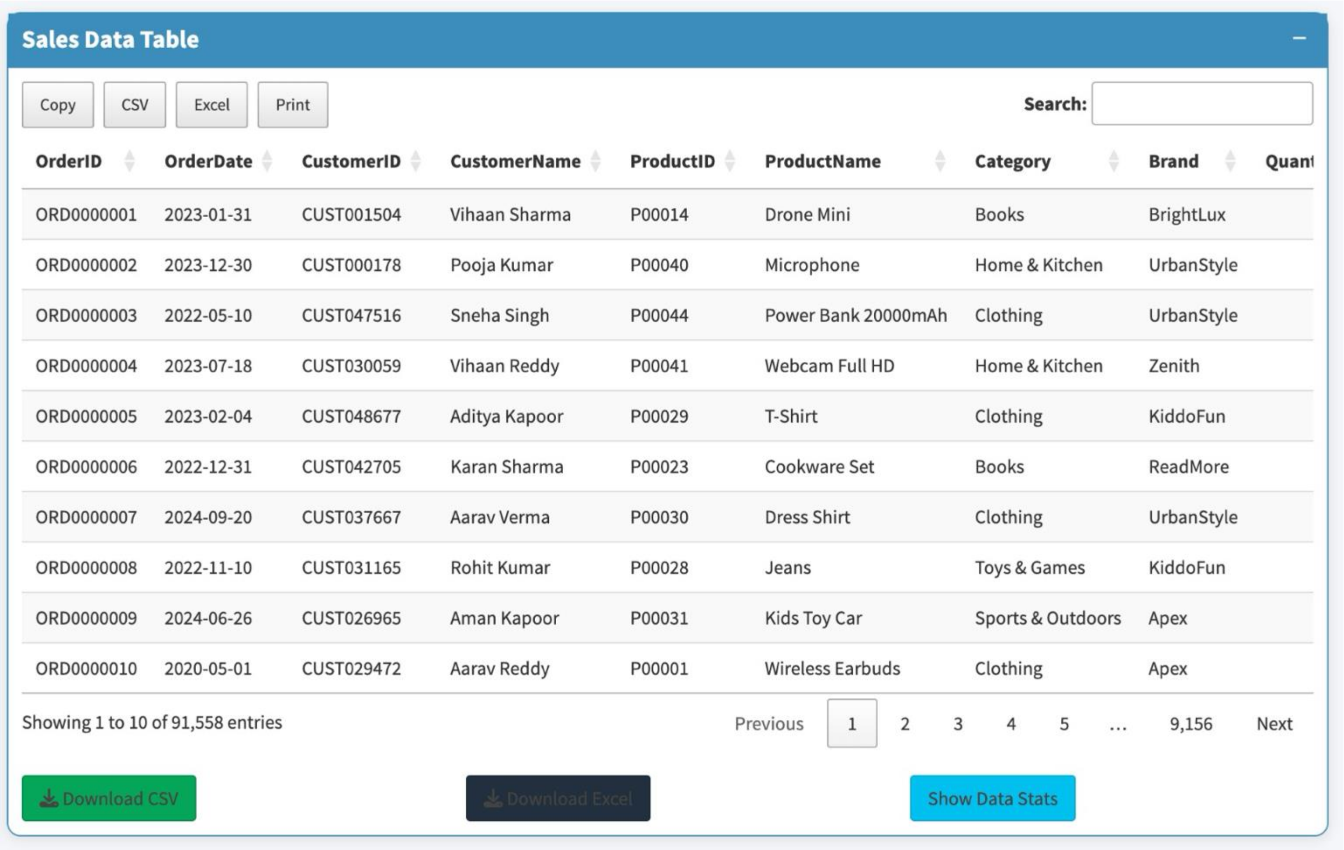
- This chart is used to monitor operational performance, showing what fraction of total orders has reached each key stage in the order lifecycle.

- **This dashboard provides a deeper analytical layer to sales data. The Scatter Plot investigates the impact of pricing strategy (discounts) on sales size and outcome. The Box Plot analyzes customer purchasing behavior and preferences based on payment method. Together, they help optimize promotions and understand payment-related sales trends.**

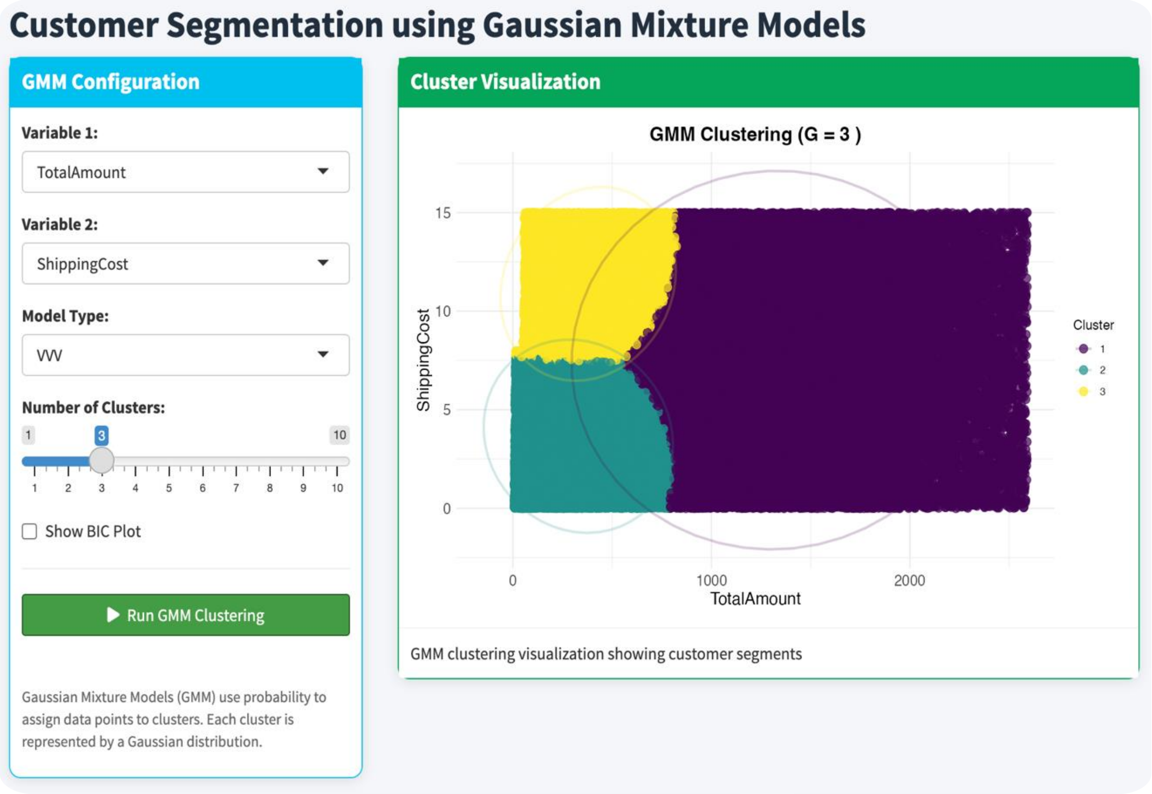


- Purpose: This table serves as the underlying data source for all the charts and dashboards in the previous screenshots. It allows for granular inspection, data verification, and detailed analysis of individual transactions.

- Summary: This is the operational sales ledger or database view, providing full access to every transaction record that powers the summary metrics and visualizations in the analytics dashboard.



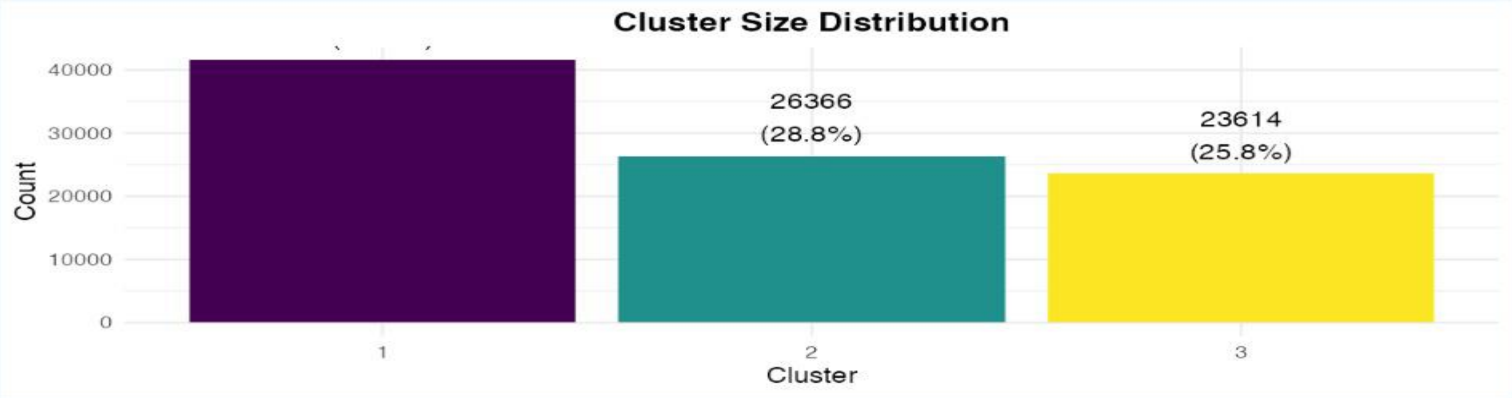
OrderID	OrderDate	CustomerID	CustomerName	ProductID	ProductName	Category	Brand	Quant
ORD0000001	2023-01-31	CUST001504	Vihaan Sharma	P00014	Drone Mini	Books	BrightLux	
ORD0000002	2023-12-30	CUST000178	Pooja Kumar	P00040	Microphone	Home & Kitchen	UrbanStyle	
ORD0000003	2022-05-10	CUST047516	Sneha Singh	P00044	Power Bank 20000mAh	Clothing	UrbanStyle	
ORD0000004	2023-07-18	CUST030059	Vihaan Reddy	P00041	Webcam Full HD	Home & Kitchen	Zenith	
ORD0000005	2023-02-04	CUST048677	Aditya Kapoor	P00029	T-Shirt	Clothing	KiddoFun	
ORD0000006	2022-12-31	CUST042705	Karan Sharma	P00023	Cookware Set	Books	ReadMore	
ORD0000007	2024-09-20	CUST037667	Aarav Verma	P00030	Dress Shirt	Clothing	UrbanStyle	
ORD0000008	2022-11-10	CUST031165	Rohit Kumar	P00028	Jeans	Toys & Games	KiddoFun	
ORD0000009	2024-06-26	CUST026965	Aman Kapoor	P00031	Kids Toy Car	Sports & Outdoors	Apex	
ORD0000010	2020-05-01	CUST029472	Aarav Reddy	P00001	Wireless Earbuds	Clothing	Apex	



- Process Summary
- The tool allows an analyst to:
- Choose key customer behavior variables.
- Configure and run a probabilistic clustering algorithm (GMM).
- Visualize the resulting customer segments to identify distinct groups, such as:
- High-spenders with high shipping costs
- Low-spenders with low shipping costs
- A potential middle group.
- Overall Purpose
- This segmentation is used for targeted marketing, personalized offers, and strategic planning by understanding different customer value and behavior profiles.

Cluster Analysis

Cluster Distribution



Cluster Characteristics

Show 10 **entries** **Search:**

Statistical Characteristics of Each Cluster

Cluster	Count	Percentage	Mean_X	SD_X	Mean_Y	SD_Y	M
1	41578	45.4	814.92	612.08	7.4	4.33	
2	26366	28.8	814.92	612.08	7.4	4.33	
3	23614	25.8	814.92	612.08	7.4	4.33	

Showing 1 to 3 of 3 entries

Previous 1 Next

GMM Model Details

```
GAUSSIAN MIXTURE MODEL SUMMARY
==============================

Model Information:
  Model Name: VVV
  Number of Components (G): 3
  Variables: TotalAmount and Shipp:
  Observations: 91558

Log-likelihood: -964086.6
BIC: -1928367
Number of Parameters: 17

Mixing Probabilities:
  Component 1: 0.491
  Component 2: 0.268
  Component 3: 0.241

Component Means:
                  [,1]   [,2]   [,3]
TotalAmount   1276.16 348.40 394.75
ShippingCost     7.52   3.69  11.28
```

Download Cluster Assignments

- Key Segment Insights:
- Cluster 1 (High-Value): Highest average spend (~$1,276) with moderate shipping cost. The most valuable segment.
- Cluster 2 (Low-Spend, Low-Cost): Low average spend (~$348) and the lowest shipping cost.
- Cluster 3 (Medium-Spend, High-Cost): Moderate average spend ($395) but the highest average shipping cost ($11.28). This could be a segment that orders heavy/bulky items or chooses faster shipping.
- 3. GMM Model Details (Technical Summary)
- Provides the technical specifications of the clustering model:
- Model Type: VVV (a fully flexible GMM where volume, shape, and orientation of clusters can vary).
- Number of Clusters (G): 3
- Observations: 91,558 (all customers/orders).
- Log-Likelihood & BIC: Statistical measures of model fit.
- Mixing Probabilities: Confirms the cluster sizes (e.g., Component/Cluster 1 has a 49.1% probability weight).
- Component Means: Matches the Mean_X and Mean_Y from the table, confirming the cluster centroids.
- Final Feature: A "Download Cluster Assignments" button to export which customer belongs to which segment for further marketing use.
- Overall Summary
- This page confirms that the customer base is successfully segmented into three distinct behavioral groups primarily defined by order value and shipping cost. It provides both a business-friendly summary (size and profile of segments) and a technical model audit, enabling actionable marketing strategies and model validation.

GMM Clustering

- Gaussian Mixture Models (GMM) Explained
- What is GMM?
- Gaussian Mixture Models (GMM) is a probabilistic clustering algorithm that assumes all data points are generated from a mixture of a finite number of Gaussian distributions with unknown parameters.
- Core Concept
- GMM is a probabilistic clustering algorithm that models data as coming from a mixture of multiple Gaussian distributions, providing soft clustering where each point has probabilities of belonging to different clusters.
- Key Features
- Three Model Categories:
- Spherical Models: Equal/unequal sized spherical clusters
- Diagonal Models: Axis-aligned ellipsoids (EEI, VEI, EVI, VVI)
- Ellipsoidal Models: Rotated ellipsoids (EEE, VEE, EVE, VVE, EEV, VEV, EVV, VVV)

```r
# Reactive for GMM results
gmm_results <- reactiveValues(
  model = NULL,
  data = NULL,
  clusters = NULL
)

# Run GMM when button is clicked
observeEvent(input$run_gmm, {
  req(input$gmm_var1, input$gmm_var2)

  # Check if selected variables exist in the dataset
  if (!(input$gmm_var1 %in% names(df_processed) && input$gmm_var2 %in% names(df_processed))) {
    showNotification("Selected variables not found in dataset", type = "error")
    return()
  }

  # Get selected data
  dat_cluster <- filtered_data()[, c(input$gmm_var1, input$gmm_var2)]
  dat_cluster <- na.omit(dat_cluster)

  if (nrow(dat_cluster) < 10) {
    showNotification("Not enough data for clustering", type = "warning")
    return()
  }

  # Run GMM
  tryCatch({
    if (input$gmm_model == "auto") {
      fit <- Mclust(dat_cluster, G = input$gmm_clusters, verbose = FALSE)
    } else {
      fit <- Mclust(dat_cluster, G = input$gmm_clusters,
                    modelNames = input$gmm_model, verbose = FALSE)
    }

    # Store results
    gmm_results$model <- fit
    gmm_results$data <- dat_cluster
    gmm_results$clusters <- fit$classification

    showNotification("GMM clustering completed successfully!", type = "success")

  }, error = function(e) {
    showNotification(paste("Error in GMM:", e$message), type = "error")
  })
})
```

- Advantages:
- Soft Clustering - Probabilistic assignments, handles overlaps
- Flexible Shapes - Models ellipsoidal clusters (not just spheres)
- Statistical Foundation - Uses EM algorithm, BIC/AIC for model selection
- Uncertainty Quantification - Provides probability measures
- Limitations:
- Computationally Intensive - Slower than K-means
- Sensitive to Initialization - May need multiple runs
- Gaussian Assumption - Works best with normally distributed data
- In Your Amazon Dashboard

- Applications:
- Customer Segmentation using purchase behavior
- Pattern Discovery in sales data
- Targeted Marketing insights
- Common Segments Found:
- High-Value Customers: High spending, premium services
- Bargain Hunters: Discount-focused, price-sensitive
- Occasional Buyers: Low frequency, small purchases

- How to Use:
- Select Variables (e.g., TotalAmount + ShippingCost)
- Choose Model Type (Start with VVV for flexibility)
- Set Cluster Count (1-10 based on BIC)
- Analyze Results - Visualize, interpret, download
- Best Practices
- Start with VVV model (most flexible)
- Check BIC plots for optimal cluster number
- Validate clusters with business context
- Normalize features if scales differ greatly
- When to Choose GMM Over Alternatives:
- . Use GMM when: Need probabilistic assignments, clusters overlap, ellipsoidal shapes
- . Consider K-means when: Need speed, spherical clusters okay
- . Consider DBSCAN when: Irregular cluster shapes, noise present
- GMM provides sophisticated, model-based clustering ideal for discovering nuanced customer segments in e-commerce data like Amazon sales, enabling data-driven business strategies.

Thanks